

RQSC JUSTIFICATION REPORT



Rolls-Royce

ALL SHADED SECTIONS ARE FOR PRODUCIBILITY / ME TEAM USE ONLY

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 1 of 10 Pages
EDNS No.:	EDNS01001274063	

1.		RQSC REGISTER		2.		REQUESTED BY	
DATE REQUESTED		23 rd June 2025		REQUESTOR'S NAME		David Rodríguez Jorge	
DATE REGISTERED				REQUESTOR'S PHONE No.		+34 944662435	
				REQUESTOR'S E-MAIL:		david.rodriguezj@itpaero.com	
MANUFACTURING ENGINEER		Joe Sullivan		David Rodríguez Jorge david.rodriguezj@itpaero.com			
MANUFACTURING ENGINEERING MANAGER		N/A					
ENGINE	RB3043-B2C	COMPONENT DESCRIPTION	VANE, LPT STAGE 4		TO BE SENT TO:		
NETWORK ACTIVITY NUMBER		APPLICABLE PART NUMBER/S	LV15218/LV15217		PART CLASSIFICATION	SIGNIFICANT	
FOR ANY INITIAL TRIAGE/DISCUSSIONS WHICH HAVE TAKEN PLACE, PLEASE IDENTIFY THE PARTICIPANTS INVOLVED AND WHICH FUNCTIONS THEY REPRESENT			Paulina Pérez Hernández – Stress José Rubén Heras Rojas – Functional David Rodríguez Jorge – Mat. & Processes		DOES THE DEFINED CHANGE IMPACT THE VISUAL APPEARANCE OF THE COMPONENT?		☐ YES ☒ NO
					HAS THE QUALITY EXECUTIVE ACCEPTED THE PROPOSED CHANGES?		☒ YES ☐ NO

3.	REASON FOR ALTERATION (SELECT AT LEAST ONE)	
3.1 COST		3.2 PROACTIVE (IMPROVEMENT)
☐ 1a. Actual recurring product cost benefit. ☐ 1b. Actual non-recurring cost benefit. ☐ 1c. Cost Avoidance.		☐ 2a. Concession avoidance. ☐ 2b. Supplier transfer.
3.3 MANUFACTURING NON-CONFORMANCE		
☐ 3a. To reflect proven manufacturing capability on non-right first-time feature(s) (RFT). ☒ 3b. Concession reduction. ☐ 3c. To reflect the agreed Feature Inspection Plan / Conformance Matrix (CM).		
3.4 ☐ Is RQSC intended for DAR? If so please add RQSS it is intended to replace in additional comments.		
3.5 ADDITIONAL COMMENTS		

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC JUSTIFICATION REPORT



Rolls-Royce

ALL SHADED SECTIONS ARE FOR PRODUCIBILITY / ME TEAM USE ONLY

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 2 of 10 Pages
EDNS No.:	EDNS01001274063	

4. BUSINESS JUSTIFICATION (QUALIFY SECTION 3) Filled in by originator using best information available.			
CRISP No: N/A		ARM No: N/A	
If other, please specify below:			
Value of RQSC relaxation (£); N/A			
Use the benefits calculator located online on InfoCentre - Producibility OR If you cannot access the link please use the benefits calculator shown below.			
Benefits Calculator: N/A			
Assumptions			
v. Standard Cost of Part		y. 12 months demand from date of implementation	
w. Working days per week		z. Exchange rate to convert to GBP	
Improvement Activity Scenarios (complete only relevant sections)			
Concession Reduction ☐ YES ☐ NO		Raw Material Saving ☐ YES ☐ NO	
a. Concession Reduction		a. Cost of the raw material per unit before	
b. Concession per unit BEFORE		b. Cost of the raw material per unit after	
c. Value of each Line item		Hard Tangible (CRISP): $(a - b) * y * z$	
d. Average number of parts per concession			
Cost Avoidance: $(b - a) * y / d * c * z$		Consumable Reduction ☐ YES ☐ NO	
		a. Total cost of consumable in the last 12 months	
		b. Expected total cost of consumable in the next 12 months	
		c. Cost of consumable reduction below budget	
		Tangible (Cost rate): $c * z$	
		Cost Avoided: $(a - b - c) * z$	
		Cycle time reduction ☐ YES ☐ NO	
		a. Actual cycle time before (hours)	
		b. Actual cycle time after (hours)	
		c. SAP cycle time reduction (hours)	
		d. FTE variable cost rate	
		Hard Tangible (CRISP): $c * d * 2 * y * z$	
		Tangible (Cash): $y / (52 * w) * v * z * (a - b) / 24$	
		Cost Avoidance: $(a - b) * d * y * z - \text{Hard Tangible}$	
		Lead Time Reduction ☐ YES ☐ NO	
		a. Numbers of concessions	
		b. Lead time saving due to concession reduction (working days)	
		c. Number of rework eliminated	
		d. Lead time saving due to rework reduction (working days)	
		Tangible (Cash): $y / (52 * w) * v * z * (a * b + c * d)$	
QUALITY:			
DELIVERY:			

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 3 of 10
EDNS No.:	EDNS01001274063	Pages

5.1 CURRENT RQSC SPECIFICATION

Requestor to provide clear indication of sections to be changed and add any tables and pictures. Multiple changes to be clearly separated.
Current RQSC issue level and amendment number used to be included.

RQSC18Q-4038 Issue 3 – Section 6.2.1:

Table 3. Allowable blendings

Zone	Depth	Extension	Feature	Remark	Remark 2
A0	None	None	TE & LE (3 mm) Rear inner platform (flowpath and no flowpath)	N/A	N/A
A1	0.4 mm	- Faired min 5X Depth, 6mm max - Separated 3X the max blend faired	Bosses	N/A	8 per cast
A2	0.381 mm deep or 25% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	LE (3 mm) Under line defined	Maximum 6 per airfoil	24 per casting The combined total depth/thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.
B1	0.381 mm deep or 25% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	Airfoil (suction and pressure surfaces including fillets as indicated) Platforms (washed surfaces)	Maximum 6 per airfoil.	Only one blend per fillet. Maximum 6 per airfoil (all B1 or 1xB2 + 5xB1) Maximum total number of blendings 24 per castings.
B2	0.75 mm deep or 50% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	Airfoil (suction and pressure surfaces indicated)	Only one blend per airfoil.	The combined total depth / thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.
C	0.5 mm deep	- Faired min 5X Depth, 4 mm max - Separated 3X the max blend faired	None gas washed surfaces as indicated	No blendings aligned on opposite sides of inner platform (flowpath and no flowpath) surfaces are allowed.	The combined total depth/thickness reduction of two blends aligned (interaction of B and C zones) on opposite sides of mid outer platform (flowpath and no flowpath) surfaces may not exceed 0.5 mm.

*LST: Lowest Section Thickness

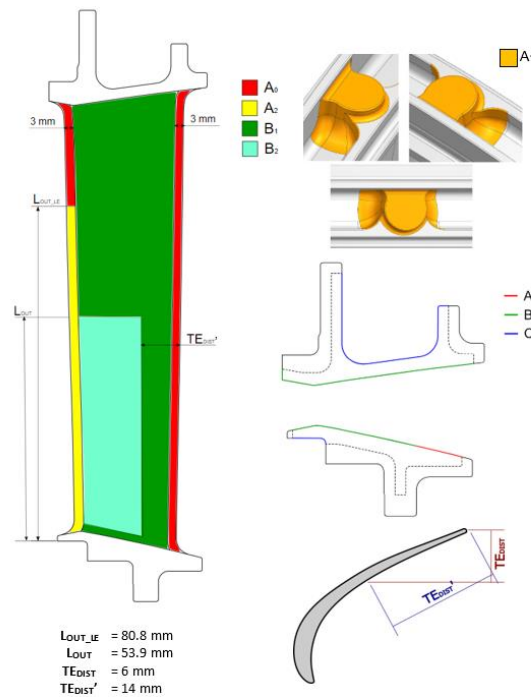


Figure 6. Blending allowance areas.

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 4 of 10
EDNS No.:	EDNS01001274063	Pages

6.1 RQSC SPECIFICATION RELAXATION REQUESTS TO BE DEFINED

Requestor to provide revised limits, tables and pictures. Multiple changes to be clearly separated and linked to items in Section 5.

RQSC18Q-4038 Issue 4 – Section 6.2.1:

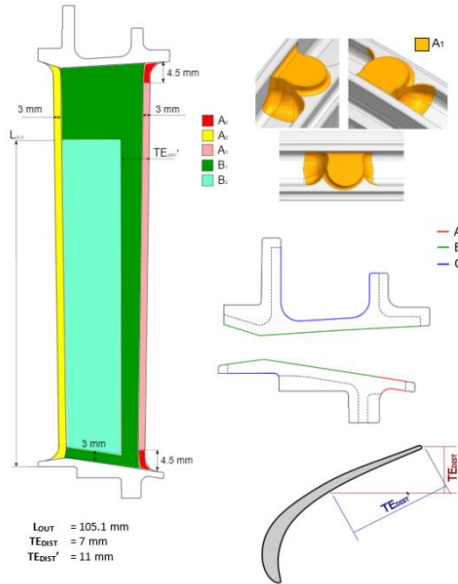


Figure 6. Defect zones (All dimensions in mm).

Table 3. Allowable blendings

Zone	Depth	Extension	Feature	Remark	Remark 2
A0	None	None	TE & LE (3 mm) Rear inner platform (flowpath and no flowpath)	N/A	N/A
A1	0.4 mm	- Faired min 5X Depth, 6mm max - Separated 3X the max blend faired	Bosses	N/A	8 per cast
A2	0.381 mm deep or 25% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	LE (3 mm) Under line defined	Maximum 6 per airfoil	Maximum 6 per airfoil (either A2 and/or A3 combination) Maximum total number of blendings 24 per castings.
A3	0.5 mm deep or up to min wall 0.535 mm	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	TE (3 mm) Under line defined	Maximum 6 per airfoil	The combined total depth/thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.
B1	0.381 mm deep or 25% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	Airfoil (suction and pressure surfaces including fillets as indicated) Platforms (washed surfaces)	Maximum 6 per airfoil.	Only one blend per fillet. Maximum 6 per airfoil (all B1 or 1xB2 + 5xB1) Maximum total number of blendings 24 per castings.
B2	0.75 mm deep or 50% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	Airfoil (suction and pressure surfaces indicated)	Only one blend per airfoil.	The combined total depth / thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.
C	0.5 mm deep	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	None gas washed surfaces as indicated	No blendings aligned on opposite sides of inner platform (flowpath and no flowpath) surfaces may not exceed 0.5 mm	The combined total depth / thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.

*LST: Lowest Section Thickness

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC JUSTIFICATION REPORT



Rolls-Royce

ALL SHADED SECTIONS ARE FOR PRODUCIBILITY / ME TEAM USE ONLY

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 5 of 10 Pages
EDNS No.:	EDNS01001274063	

7.1	SUPPORTING EVIDENCE <i>Requestor is to capture relevant history (deviations from the current quality standard), as well as root causes and any corrective actions (if any) that has been put in place prior. Examples include Concessions and Deviation Permits approved for similar relaxations (fully approved documents together with specialist task list), presentations shared with specialists, capability data charts, etc.</i>
See point 9 for details.	
8.1	ALTERATION INSTRUCTION <i>To be compiled by Design after specialist responses are captured - "Accepted, Accepted not as requested, Rejected". Value, tables, and illustrations should be shown in the final accepted condition.</i>
Accepted: ☒ Accepted not as requested: ☐ Rejected: ☐	
If the amendment is to be accepted as specified in section 6, section 8 may be endorsed 'complete as per section 6.#' and otherwise left blank. If the amendment is to be accepted not as requested, the alteration should be detailed below: COMPLETE AS PER SECTION 6	

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 6 of 10
EDNS No.:	EDNS01001274063	Pages

5.2 CURRENT RQSC SPECIFICATION

Requestor to provide clear indication of sections to be changed and add any tables and pictures. Multiple changes to be clearly separated.
Current RQSC issue level and amendment number used to be included.

RQSC18Q-4038 Issue 3 – Section 6.2.2:

Table 4. Allowable cutbacks.

Zone	Max Depth	Max Length	Quantity	Remark
Leading Edge	1 mm	10 mm	4 x NGV Ring Max 1 x Airfoil	Defects must be considered altogether. The deepest and longest defect determines the cutback allowance in terms of quantity per NGV Ring.
	2 mm	10 mm	1 x NGV Ring	
Trailing Edge	1 mm	5 mm	12 x NGV Ring Max 1 x Airfoil	
		10 mm	6 x NGV Ring Max 1 x Airfoil	
		20 mm	3 x NGV Ring Max 1 x Airfoil	
	2 mm	10 mm	2 x NGV Ring Max 1 x Airfoil	
		20 mm	1 x NGV Ring Max 1 x Airfoil	

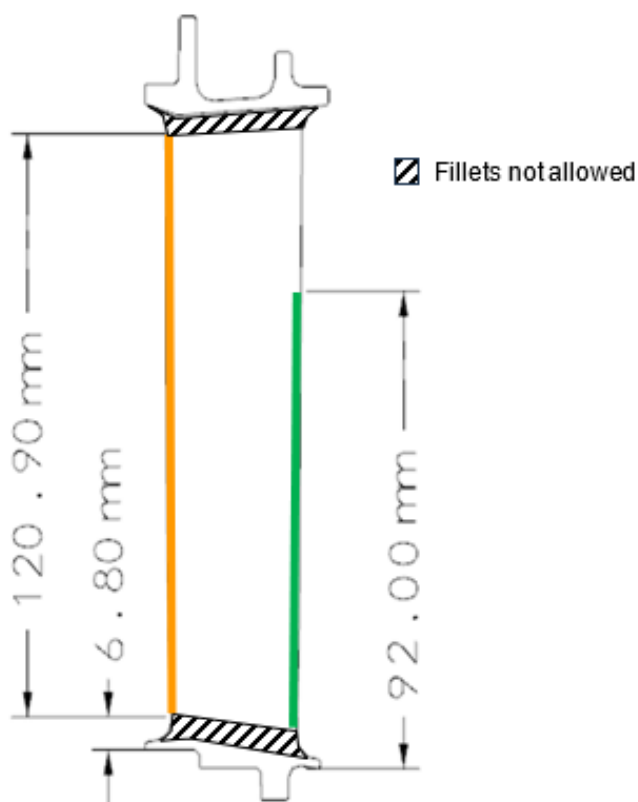


Figure 8. Cutbacks allowance areas.

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 7 of 10
EDNS No.:	EDNS01001274063	Pages

6.2 RQSC SPECIFICATION RELAXATION REQUESTS TO BE DEFINED

Requestor to provide revised limits, tables and pictures. Multiple changes to be clearly separated and linked to items in Section 5.

RQSC18Q-4038 Issue 4 – Section 6.2.2:

Table 4. Allowable cutbacks.

Zone	Max Depth	Max Length	Quantity	Remark	Remark 2
Leading Edge	1 mm	10 mm	4 x NGV Ring Max 1 x Airfoil	N/A	Defects must be considered altogether. The deepest and longest defect determines the cutback allowance in terms of quantity per NGV Ring. Min acceptable length is 4X (X=depth).
	2 mm	10 mm	1 x NGV Ring		
Trailing Edge	1 mm	5 mm	12 x NGV Ring Max 1 x Airfoil	Permitted in all the airfoil span (continuous green line + green dotted line). See Fig 8. Fillets not allowed.	
		10 mm	6 x NGV Ring Max 1 x Airfoil		
		20 mm	3 x NGV Ring Max 1 x Airfoil		
		55 mm	1 x NGV Ring Max 1 x Airfoil		
	2 mm	10 mm	2 x NGV Ring Max 1 x Airfoil	Permitted up to 92 mm of the airfoil span (green dotted line). See Fig 8. Fillets not allowed.	
		20 mm	1 x NGV Ring Max 1 x Airfoil		

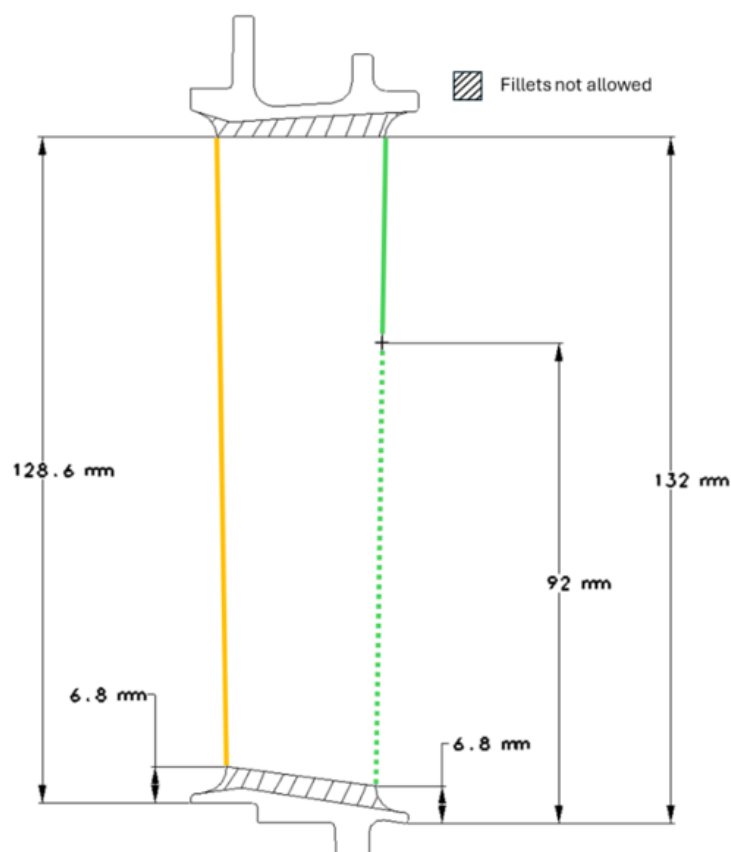


Figure 8. Cutbacks allowance areas.

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

ALL SHADED SECTIONS ARE FOR PRODUCIBILITY / ME TEAM USE ONLY

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 8 of 10 Pages
EDNS No.:	EDNS01001274063	

7.2	SUPPORTING EVIDENCE <i>Requestor is to capture relevant history (deviations from the current quality standard), as well as root causes and any corrective actions (if any) that has been put in place prior. Examples include Concessions and Deviation Permits approved for similar relaxations (fully approved documents together with specialist task list), presentations shared with specialists, capability data charts, etc.</i>
See point 9 for details.	
8.2	ALTERATION INSTRUCTION <i>To be compiled by Design after specialist responses are captured - "Accepted, Accepted not as requested, Rejected". Value, tables, and illustrations should be shown in the final accepted condition.</i>
Accepted: ☒ Accepted not as requested: ☐ Rejected: ☐	
If the amendment is to be accepted as specified in section 6, section 8 may be endorsed 'complete as per section 6.#' and otherwise left blank. If the amendment is to be accepted not as requested, the alteration should be detailed below:	
COMPLETE AS PER SECTION 6	

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC JUSTIFICATION REPORT



Rolls-Royce

ALL SHADED SECTIONS ARE FOR PRODUCIBILITY / ME TEAM USE ONLY

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 9 of 10 Pages
EDNS No.:	EDNS01001274063	

9.	ENGINEERING TECHNICAL JUSTIFICATION	<i>Specialists to record detail of justification / modification requests and any other associated data etc. in this section. Referral to specialist technical assessment sheets is an acceptable practice providing that a traceable identifier is provided.</i>
RQSC18Q-4038 has been proposed to 4th issue by ITP Aero. CSH-00035664 (Communication Sheet) has been raised as a support document which gathers the rationale given by the different disciplines for the changes introduced to the standard. As stated in the aforementioned document, references below to ITP Aero Specialists Technical Assessments support from a technical point of view this update: CSH-00034802/001. P700 Adv2 LPT NGV Rings Blendings Allowance Update, Luis Bolaños, June 14th 2025. CSH-00034813/001. Pearl 700 and Pearl 10X NGV Ring Stg 4 Additional Trailing Edge Cutbacks Acceptance, Paulina Pérez, Josue Mendonza, June 13th 2025.		

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	

RQSC JUSTIFICATION REPORT



Rolls-Royce

ALL SHADED SECTIONS ARE FOR PRODUCIBILITY / ME TEAM USE ONLY

RQSC No.:	RQSC18Q-4038 - Issue 3 to Issue 4	Page 10 of 10 Pages
EDNS No.:	EDNS01001274063	

10.	ENGINEERING APPROVAL – Design to approve the justification report and confirm that the defined alteration instruction(s) meet all specialist inputs.		
	NAME	SIGNATURE	DATE
	Sergio Benneth Díaz Tapia Eduardo Mendieta López		24.June.2025
DOES THE DEFINED CHANGE IMPACT THE VISUAL APPEARANCE OF THE COMPONENT?			☐ YES ☒ NO
DOES THE REQUESTED CHANGE AFFECT ANY RELEVANT SERVICE EXPERIENCE, INCLUDING SARS / RED TOPS?			☐ YES ☒ NO
HAS LIFECYCLE ENGINEERING IDENTIFIED ANY CHANGES REQUIRED FOR ANY ASSOCIATED OVERHAUL INSTRUCTIONS, INSPECTIONS AND/OR ENGINE MANUALS?			☐ YES ☒ NO ☐ N/A

11.	MANUFACTURING APPROVAL – Manufacturing to provide acceptance of the alteration(s) defined within section 8.		
	NAME	SIGNATURE	DATE
	Joe Sullivan		9-July-2025
Accept			
WHERE IT HAS BEEN IDENTIFIED THAT CHANGES ARE REQUIRED FOR ASSOCIATED OVERHAUL INSTRUCTIONS, INSPECTIONS AND/OR ENGINE MANUALS, IT IS THE RESPONSIBILITY OF MANUFACTURING ENGINEERING TO NOTIFY LIFECYCLE ENGINEERING OF THE UP-ISSUED RQSC.			

Jurisdiction	Export Control Classification (Rating)	Date
UK	NOT LISTED	23 June 2025
US	NOT LISTED	